is derived from the Greek word *tomos* means slice and *graphe* means to write. It is a medical imaging technology employing tomography, were digital geometry processing is used to generate a three dimensional image of the internals of an object from a large series of two dimensional X ray images taken around a single axis of rotation (Figure12.8).

CT produces volumes of data which can be manipulated through a process known as windowing in order to demonstrate various structures based on their ability to block the X ray beam.



Figure 12.8 CT Scan

Clinical significance

- Gives a clear image of bone, soft tissues and blood vessels
- Helps in the diagnosis of injuries of the inner ears and sinuses
- To detect cancer, heart and lung disorders
- For diagnosis of spinal problems and skeletal injuries
- Helps to measure bone mineral density
- To detect stroke causing clots and hemorrhage in the brain.

Positron Emission Tomographic Scanning (PET)

PET is also computerized imaging technique unlike CT. Positron emission tomography (PET) is a nuclear medicine procedure based on the measurement of positron emission from radiolabelled tracer molecules. These radiotracers allow biological processes to be measured and whole body images to be obtained which demonstrates sites of radiotracer accumulation. A PET image gives quantitative regional information on the metabolic and physiological processes. PET uses positron emitting radio isotopes (^{11}C ^{13}N ^{15}O ^{18}F) which are generated by the cyclotron. The most common radiotracer in use today is 18F-fluorodeoxyglucose (18F-FDG) which is a radio labelled sugar (glucose) molecule. These atoms are then incorporated by chemical methods into biological molecules like glucose, amino acids and ammonia. These positron emitting compounds are then injected in very small amounts into or inhaled by experimental animals or human subjects. The three dimensional distribution of the labeled trace is then probed by powerful PET cameras and the images are reconstructed by a computer. The quantitative interpretation of the image is done by varying mathematical models. They deal with the process of uptake and metabolism of the radioisotope.

Clinical significance

PET imaging is effectively used in the measurement of regional cerebral blood volume, blood flow, metabolic rates for glucose and oxygen in humans.

MRI-Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) is a non-invasive medical test that physicians use to diagnose medical conditions.

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